Construction and Physical-Space Digital Twins: Landscape

Research date: 2026-10-18. Vendor figures are self-reported. ***[uncertain]*** *= conflicting or unverified.*

1. Executive summary

- **The market is consolidating fast.**
  - Hexagon bought Avvir and owns OxBlue.
  - DroneDeploy absorbed StructionSite (brand shut down Nov 2024).
  - OpenSpace acquired Disperse (Oct 2025).
  - AMETEK bought FARO/HoloBuilder (~$920M, Jul 2025).
  - CoStar bought Matterport ($1.6B, Feb 2025).
  - Bentley bought Cesium (Sep 2024).
- **The commercial core is progress tracking against BIM and the schedule.**
  - Capture is mostly a weekly 360° helmet walk, plus drones for exteriors and laser scanners for QA.
  - Buildots raised a $130M Series E (Sep 2026) at a valuation of nearly $1B.
- **Data centers are the leading vertical:** MEP-heavy work, tight schedules, and pay applications tied to installed work.
- **The frontier is autonomous capture plus AI agents.** Examples: DroneDeploy docked drones, Spot/Unitree robots and Capture Agent robotics-as-a-service (Jul 2026); OpenSpace agents; Bentley Copilot; NVIDIA Omniverse blueprints.
- **Twins are still periodic snapshots.** Live signals (cameras, crane hooks, wearables, telematics) sit in separate silos and are rarely fused into one spatial model.

2. Construction reality capture and progress tracking

AI progress-tracking platforms

- **OpenSpace**
  - Capture: 360° helmet camera with SLAM-like auto-placement on floor plans, plus Air (drones), scans and BIM+.
  - Analytics: Track (from the Disperse acquisition) uses computer vision plus human verification and claims 95% accuracy. It connects to P6, Asta and MS Project. Waypoint 2026 added Autolocation 2.0, Site Mode, agents and a Track API.
  - Funding and pricing: ~$199M raised ($902M valuation in 2022); pricing starts around a $10k/yr minimum, scaled by volume.
- **Buildots**
  - Capture: helmet 360° walk 1-2x per week.
  - Analytics: element-level comparison against BIM and schedule across 80+ stages, plus delay forecasting. Needs clean BIM.
  - Funding and customers: $297M total and a valuation of nearly $1B (Construction Dive). Customers include Intel, Digital Realty, JE Dunn, Mortenson and HOCHTIEF.
- **Doxel**

    - Capture: started with a LiDAR robot/cart; now a 360° helmet camera ([uncertain] whether LiDAR is still offered).
    - Analytics: work-in-place per trade and system, 4D views. Focus on data centers.
    - ~$56.5M raised.
- **Track3D**
    - Capture: any 360° camera, plus scans.
    - Analytics: planned vs actual progress.
    - **Pricing: $0.15/sq ft for progress tracking, documentation free** (PR). This is the clearest sign that capture is a commodity and analytics is the paid product.
    - $14.3M raised, 400+ projects.
- **Cupix**
    - Capture: 360° video turned into a 3D map. VideoDrive captures from a vehicle at up to 40 km/h.
    - Analytics: SiteInsights for percent complete and BIM deviation.
- **Reconstruct**
    - Turns drone, 360° and phone footage into an as-built plan and 3D twin aligned with P6.
    - Example: a 600k sq ft project twinned in 2 days, versus 2+ months by scanning.
- **Avvir:** scan-vs-BIM checks; now part of Hexagon.

Aerial and robotic capture

- **DroneDeploy**
    - More than 20 trillion sq ft of imagery captured.
    - 13k docked-drone missions and 16k ground-robot missions year-to-date by Apr 2026, on Spot and Unitree Go2-W.
    - **Capture Agent** (Jul 2026, early access): the robot navigates from drawings only, with no markers, and scans after hours. Sold as robotics-as-a-service.
    - Progress, Safety and Inspection AI agents.
    - ~$349-649/mo, then enterprise pricing (Jul 2026 release).
- **Propeller Aero:** drone surveying with AeroPoints smart GCPs; volumes and cut/fill. **DirtMate** links equipment telemetry to the terrain model. ~$12k per drone per year.
- **Skydio:** X10 + Dock for scheduled autonomous missions; 3D Scan for inspection-grade models.
- **Boston Dynamics Spot:** Orbit 5.0 fleet software; Leica BLK ARC for autonomous scanning.
- **Dusty Robotics:** layout printing from the model onto the slab (1/16"), 100M+ sq ft. **Civ Robotics:** layout robots.
- **Versatile CraneView:** a crane-hook sensor and camera tracking lifts and utilization (+28% claimed); $100M+ raised.

Incumbent platforms

- **Hexagon/Leica:** BLK2GO, BLK ARC, RTC360, Reality Cloud, OxBlue cameras, Avvir.
- **Trimble:** X9 scanner, Reality Capture in Connect (accepts iPhone LiDAR), SiteVision AR.
- **Bentley:** iTwin (Capture now produces splats), SYNCHRO+ 4D with Copilot agents, Cesium / 3D Tiles.
- **Autodesk:** Construction Cloud, possibly renamed **Forma** [uncertain]. Tandem is the operations twin (BIM handover + IoT).
- **NVIDIA Omniverse:** Mega and DSX blueprints (AI-factory twins built before construction). A horizontal enabler, not a site product.
- **Unity/Unreal:** visualization runtimes used by integrators.

Live-site video and safety

Intenseye (~$90M raised), Everguard (cameras + wearables + RTLS), Evercam (cameras + drone + BIM timeline), viAct. All are detection-only and not registered to a twin.

3. General-place twins

- **Matterport (CoStar):** 14M+ spaces; FY2024 revenue $169.7M.
- **NavVis:** VLX 4, CLX and the IVION platform; $85M Series D (Aug 2026) pitched as a "data foundation for physical AI"; 2B+ m² captured.
- **Niantic Spatial:** $250M spinout; Large Geospatial Model; Scaniverse produces splats and VPS maps; VPS 2.0 gives near-cm 6DoF localization.
- **Google and Apple:** Google Photorealistic 3D Tiles as a base layer; Apple RoomPlan and Object Capture.
- **Splat tools:** Polycam (room-scale); Luma, which has pivoted to world models ($900M raised). Splat standards are settling: KHR_gaussian_splatting and a native OpenUSD 26.03 schema.
- **Smart-city twins:** Virtual Singapore, Helsinki, Rotterdam. Slow to refresh and focused on planning.

4. Academic and open-source references

- **Scan-to-BIM:** BIMNet (IFC scan-to-BIM benchmark), CV4AEC Scan-to-BIM.
- **SLAM and scene benchmarks:** Hilti SLAM Challenge, ConSLAM, S3DIS.
- **Industrial:** Industrial3D (MEP point clouds).
- **Robot-ready site twins:** BIM2RDT.
- **Splat twins for simulation:** ArtiTwinSplat (Isaac Sim-ready twins).
- **Key gap:** LiBiDT, Sci. Reports 2026 notes that **no public time-stamped, multi-capture dataset of a site under active construction exists**.

5. Comparison table

Company	Capture	Cadence	Analytics	Pricing	Funding
OpenSpace	360° helmet, drone, scans	Weekly	Auto-placement, Track, agents	~$10k/yr minimum	~$199M
Buildots	360° helmet, drone	1-2x/week	Element-level BIM progress, delay forecasting	Enterprise	$297M, ~$1B valuation
Doxel	360° (formerly LiDAR robot)	Weekly	Work in place, 4D	Premium	~$56.5M
Track3D	Any 360° camera	~6 h turnaround	Planned vs actual	$0.15/sq ft	$14.3M
Cupix	360° video, scans, drone	Per capture	Percent complete, deviation	Quote	Private

Company	Capture	Cadence	Analytics	Pricing	Funding
Reconstruct	Drone, 360°, phone	Per capture	As-built + P6	Quote	Strategic round
DroneDeploy	Drone, dock, Spot/Unitree	Autonomous, up to nightly	AI agents	$349-649/mo and up; RaaS	Private
Propeller	Drone + AeroPoints + machine sensors	Per flight	Volumes, equipment paths	~$12k/drone/yr	~$32M
Hexagon/Leica	BLK2GO/ARC, RTC360, OxBlue	Per scan	Registration, BIM checks	Hardware + cloud	Public
Bentley	iTwin Capture, Cesium	Per capture	4D, agents	Licensing	Public
Autodesk Tandem	BIM + IoT	Live (operations)	Operations twin	Subscription	Public
NavVis	Mobile scanners	Per scan	IVION	Hardware + SaaS	$85M Series D
Niantic Spatial	Phone, 360°, drone	Crowd-sourced	VPS, LGM	API	$250M
Versatile	Crane hook	Live	Lifts, utilization	Subscription	$100M+

6. Business patterns

1. **Capture is a commodity; analytics is the product.** Verified progress tied to pay applications and the schedule is what customers pay for.
2. **Pricing is per square foot or by project volume**, sold to general contractors and increasingly to owners.
3. **Human-in-the-loop verification** makes up for weak computer vision.
4. **Integrations are table stakes** (Procore, ACC/Forma, P6, Aconex), and incumbents buy point solutions.
5. **"Physical AI data" pitch:** spatial datasets sold as training data for robots and world models.
6. **Hardware-agnostic vendors vs hardware incumbents** who lock customers into their cloud.

7. Gaps and opportunities

1. **Periodic, not live.** Opportunity: one continuously updated 4D twin that fuses fixed cameras, robots, telematics and worker location, with change detection between full captures.
2. **Autonomous capture is early and fragile.** Sites change daily and break robot routes. Opportunity: robots that navigate from drawings/BIM and update their own maps; cheap wheeled or quadruped platforms; robotics-as-a-service.

3. **MEP and hidden work gets covered by drywall before it is captured.** Opportunity: triggered capture before close-in.
4. **Trust and accuracy are the number one adoption barrier**, and vendor claims are unaudited. Opportunity: element-level confidence scores, provenance and an open benchmark.
5. **Dependence on clean BIM.** Opportunity: progress tracking driven by drawings or the schedule; automatic scan-to-BIM.
6. **Civil, solar, earthworks and mining are underserved.**
7. **Safety, equipment and worker data are not registered to the twin.**
8. **Construction-to-operations handover is disconnected.**
9. **Splats look real but carry no meaning or measurement guarantees.** Opportunity: semantic, measurable splats linked to BIM elements.

Where a newcomer can differentiate:

- (a) Near-real-time updates from fixed cameras and robots.
- (b) Autonomous robotic capture sold as a service.
- (c) Equipment and worker tracking fused into the same spatial model.
- (d) Verifiable BIM-vs-as-built deviation with confidence scores.
- (e) Underserved verticals and projects without BIM.